Homological Algebra (Extension of Sheaves) :

(Vistoli's Notes) (§3)

→ Part - I : Extensions of sheaves of $\mathcal{O}_X$-modules.

Their properties viz. pullback, pushforward, sum etc.

→ Part - II : Extension cocycles and group of extension classes. Relation b/w the extension classes of $\mathcal{F}$ by $\mathcal{G}$ and $\mathrm{Ext}^2_{\mathcal{O}}(\mathcal{F},\mathcal{G})$.

(Lec. 6 Abstract lifting)

→ Part - III : Generalization to $\mathcal{F}^\bullet$ from $\mathcal{F}$ and an analogous theorem.

(Lec 7. Lifting of l.c.i. morphism)

Given $A' \to A = A'/\mathfrak{a}$; $\mathfrak{m}_{A'}\mathfrak{a} = 0$ (A': Noetherian local ring)

a scheme M' (flat) over A' and a scheme X over $M (= M'|_{\mathrm{Spec}\,A})$;

X' flat is an abstract lifting s.t. $X \hookrightarrow X'$ and $X'|_{\mathrm{Spec}\,A} \cong X$.

$\downarrow$

M'

Given lci morphism $X \xrightarrow{f} M$; its lifting (X', f') X': an abstract lifting of X and $f' : X' \to M'$ s.t. $f'|_X = f$.

if $f_0 : X_0 \to M_0$; $\mathcal{F}^\bullet = [\cdots \to 0 \to \underset{\deg -1}{f_0^* \Omega_{M_0/k}} \to \underset{\deg 0}{\Omega_{X_0/k}} \to 0]$

$\mathcal{G} = \mathcal{O}_X$ (deg 0).

Part - I : X : scheme

All sheaves are sheaves of $\mathcal{O}_X$-modules.

Given $\mathcal{F}, \mathcal{G}$ sheaves ; an extension of $\mathcal{F}$ by $\mathcal{G}$

(*) — $0 \to \mathcal{G} \xrightarrow{i_\mathcal{E}} \mathcal{E} \xrightarrow{k_\mathcal{E}} \mathcal{F} \to 0$ (a short exact sequence).

$\mathrm{Ext}^1_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$: Choose inj. sheaf $\mathcal{G} \overset{i}{\hookrightarrow} \mathcal{I}$ and

$$\mathrm{Ext}^1_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) := \mathrm{Hom}(\mathcal{F}, \mathcal{I}/\mathcal{G}) \,/\, \mathrm{Im}\left(\mathrm{Hom}(\mathcal{F}, \mathcal{G}) \to \mathrm{Hom}(\mathcal{F}, \mathcal{I})\right)$$

Given a ses (*) ; $0 \to \mathcal{G} \to \mathcal{E} \to \mathcal{F} \to 0$

$0 \to \mathcal{I}$

$\mathrm{Hom}(\mathcal{E}, \mathcal{I}) \to \mathrm{Hom}(\mathcal{G}, \mathcal{I})$

$i' \longmapsto i$

$\bar{i'} : \mathcal{E}/\mathcal{G} \to \mathcal{I}/\mathcal{G}$

$\mathcal{F}$ $[\bar{i'}]$

Conversely given $[\phi] \in \mathrm{Ext}^1_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$

$\psi : \mathcal{F} \to \mathcal{I}/\mathcal{G}$; pullback $\{(f, e) \in \mathcal{F} \oplus \mathcal{I} \mid \psi(f) = e \pmod{i(\mathcal{G})}\}$

$$\begin{array}{ccccccc} 0 & \to & \mathcal{G} & \to & \phi^*\mathcal{I} & \to & \mathcal{F} \to 0 \\ & & \| & & \vdots & & \downarrow \phi \\ 0 & \to & \mathcal{G} & \to & \mathcal{I} & \to & \mathcal{I}/\mathcal{G} \to 0 \end{array}$$

$\downarrow$ e.

- $\phi' = \phi + i \circ \psi \quad (\psi : \mathcal{F} \to \mathcal{G})$

$$\phi^*\mathcal{A} = \{(f', e') \in F \oplus \mathcal{G} \mid \phi'(f') = e' \pmod{i(\mathcal{G})}\}$$

$$\downarrow \simeq \qquad \downarrow \qquad\qquad \phi(f') + (\psi \circ i)(f')$$

$$\phi^*\mathcal{G} \ni (f', e') \qquad \Rightarrow (\phi' - \phi)(f') = (\psi \circ i)(f') = 0 \pmod{i(\mathcal{G})}$$

$\Rightarrow$ bijection $\mathrm{Ext}^1_{\mathcal{O}}(F, \mathcal{G}) \xrightarrow{\sim} \{\text{isomorphism classes of } \mathcal{F} \text{ by } \mathcal{G}\}$

pullback: Given an extension $0 \to \mathcal{G} \to \Sigma \to F \to 0$

and a map $f: F' \to F$;

$$\begin{array}{ccccccc} & & & i & & & \\ 0 & \to & \mathcal{G} & \to & f^*\Sigma & \to & F' \to 0 \\ & & \| & & \downarrow \psi & & \downarrow f \\ 0 & \to & \mathcal{G} & \xrightarrow{i_\Sigma} & \Sigma & \to & F \to 0 \\ & & & & \downarrow k_\Sigma & & \end{array}$$

$$f^*\Sigma = \{(e, x') \in \Sigma \oplus F' \mid f(x') = k_\Sigma(e)\}$$

$i_{f^*\Sigma}: \mathcal{G} \to f^*\Sigma$ is given by

$$y \mapsto (0, i_\Sigma(y))$$

universal property: Given any extension $\mathcal{E}$ of F by $\mathcal{G}$

and a commutative diagram

$$\begin{array}{ccccccc} 0 & \to & \mathcal{G} & \to & \mathcal{E}' & \to & F' \to 0 \\ & & \| & & \downarrow \psi' & & \downarrow f \\ 0 & \to & \mathcal{G} & \to & \Sigma & \to & F \to 0 \end{array}$$

$\exists\, f^*\Sigma \xrightarrow{\phi} \mathcal{E}'$ s.t.

$\psi'\phi = \psi$.

push forward : Given an extension $\mathcal{E}$ of $\mathcal{F}$ by $\mathcal{G}$ and $g: \mathcal{G} \to \mathcal{G}'$;

$$\begin{array}{ccccccc} 0 & \to & \mathcal{G} & \xrightarrow{i_{\mathcal{E}}} & \mathcal{E} & \xrightarrow{k_{\mathcal{E}}} & \mathcal{F} \to 0 \\ & & \downarrow g & & \downarrow \psi & & \| \\ 0 & \to & \mathcal{G}' & \to & g_*\mathcal{E} & \to & \mathcal{F} \to 0 \end{array}$$

$$g_*\mathcal{E} := \{(e, \lambda') \in \mathcal{E} \oplus \mathcal{G}' \} / \sim$$

$$(\forall\, \lambda \in \mathcal{G}) \quad (i_{\mathcal{E}}(\lambda), -g(\lambda)) \sim (0, i'(\lambda))$$

$$\lambda' \mapsto (0, \lambda')$$

$$\mathcal{G}' \to g_*\mathcal{E} \to \mathcal{F}$$

$$(e, \lambda') \mapsto k_{\mathcal{E}}(e) \qquad k_{\mathcal{E}}(e + i_{\mathcal{E}}(\lambda)) = k_{\mathcal{E}}(e).$$

- universal property : If there is a commutative diagram

$$\begin{array}{ccccccc} 0 & \to & \mathcal{G} & \to & \mathcal{E} & \to & \mathcal{F} \to 0 \\ & & \downarrow g & & \downarrow \psi' & & \| \\ 0 & \to & \mathcal{G}' & \to & \mathcal{E}' & \to & \mathcal{F} \to 0 \end{array}$$

then $\exists\, \psi : g_*\mathcal{E} \xrightarrow{\sim} \mathcal{E}'$ s.t. $\psi' \phi = \psi$

Sum of two extensions: Given extensions

$$0 \to \mathcal{G} \xrightarrow[i_{\xi_i}]{} \xi_i \xrightarrow[k_{\xi_i}]{} \mathcal{F} \to 0 \quad ; \text{ their sum is defined}$$

$$\begin{array}{ccccccccc}
0 & \to & \mathcal{G} & \to & (+)_* \Delta^*(\xi_1 \oplus \xi_2) & \to & \mathcal{F} & \to & 0 \\
 & & (+)\uparrow & & \uparrow & & \| & & \\
0 & \to & \mathcal{G}\oplus\mathcal{G} & \to & \Delta^*(\xi_1 \oplus \xi_2) & \to & \mathcal{F} & \to & 0 \\
 & & \| & & \downarrow & & \downarrow \Delta & & \\
0 & \to & \mathcal{G}\oplus\mathcal{G} & \to & \xi_1 \oplus \xi_2 & \to & \mathcal{F}\oplus\mathcal{F} & \to & 0
\end{array}$$

$$\xi_1 + \xi_2 := (+)_* \Delta^*(\xi_1 \oplus \xi_2).$$

section: $\{(e_1, e_2) \in \xi_1 \oplus \xi_2 \mid k_1(e_1) = k_2(e_2)\} / \{(i_1(g_1), i_2(g_2)) \mid g_1 + g_2 = 0\}$.

$$\begin{array}{ccc}
\mathcal{G} & \to & \xi_1 + \xi_2 \\
i \downarrow & \nearrow & \downarrow [e_1, e_2] \\
\mathcal{G} & & i'(e_1) + i''(e_2)
\end{array}
\qquad
\begin{array}{ccc}
g_1 & \to & [i_1(g_1), 0] \\
i \downarrow & & \downarrow \\
 & & i'(e_1)
\end{array}$$

$$\begin{array}{ccc}
\mathcal{G} & \to & \xi_1 \\
i \downarrow & \nearrow & i' \text{ \& } i'' \\
\mathcal{G} & &
\end{array}$$

$$\xi_1 \to [\tilde{i'}]$$

$$\xi_2 \to [\tilde{i''}].$$

$$\xi_1 + \xi_2 \to [\tilde{i'} + \tilde{i''}]$$

$$\tilde{i'} : \mathcal{F} \to \mathcal{I}/\mathcal{G}$$

$$\tilde{i''} : \mathcal{F} \to \mathcal{I}/\mathcal{G}.$$

Properties: If $\mathcal{E}_i$, $i=1,2$ extensions of F by $\mathfrak{h}$ and $g_1, g_2, g: \mathfrak{h} \to \mathfrak{h}'$, a homomorphism then

i) $g_*(\mathcal{E}_1 + \mathcal{E}_2) \simeq g_*\mathcal{E}_1 + g_*\mathcal{E}_2$

ii) For $\mathbf{0}: \mathfrak{h} \to \mathfrak{h}'$; $\mathbf{0}_*\mathcal{E} = \mathbf{0}_{F,\mathfrak{h}}$ the split extension

iii) $(g_1+g_2)_*\mathcal{E} \simeq g_{1*}\mathcal{E} + g_{2*}\mathcal{E}$

pf: i) $g_*(+)_*\circ\Delta^*(\mathcal{E}_1 \oplus \mathcal{E}_2)$

$\cong (g\circ(+))_*\circ\Delta^*(\mathcal{E}_1\oplus\mathcal{E}_2)$

$= (+)_*\circ(g\oplus g)_*\Delta^*(\mathcal{E}_1\oplus\mathcal{E}_2)$

$= (+)_*\circ\Delta^*(g\oplus g)^*(\mathcal{E}_1\oplus\mathcal{E}_2)$

$= (+)_*\circ\Delta^*(g^*\mathcal{E}_1 \oplus g^*\mathcal{E}_2)$

$= g_*\mathcal{E}_1 + g_*\mathcal{E}_2$.

ii)
$$\begin{array}{ccccc} 0 & \to & \mathfrak{h} & \xrightarrow{i_1} & \mathcal{E} \\ & & \downarrow 0 & & \downarrow \\ 0 & \to & \mathfrak{h}' & \to & 0_*\mathcal{E} = \{(e,\mathfrak{y}') \mid e\in\mathcal{E}, \mathfrak{y}'\in\mathfrak{h}'\}/\sim \end{array}$$

$\mathfrak{y}' \mapsto [e,\mathfrak{y}']$ $\quad (e,\mathfrak{y}') \sim (e+i_1(g),\mathfrak{y}')$ $(\forall g\in G)$

• Notations: i) $\mathcal{E}_1 - \mathcal{E}_2 := \mathcal{E}_1 + (-\mathcal{E}_2)$ where

$$0 \to \mathcal{U}_\mathcal{F} \xrightarrow{-i} -\mathcal{E}_2 \xrightarrow{k} \mathcal{F} \to 0$$

ii) For a morphism $f: \mathcal{E}_1 \to \mathcal{E}_2$ of extensions of $\mathcal{F}$ by $\mathcal{U}_\mathcal{F}$; $\Theta f: -\mathcal{E}_1 \to -\mathcal{E}_2$ is denoted although by definition it is the map f.

iii) $\chi_{\mathcal{E}_1, \mathcal{E}_2} : \mathcal{E}_1 + \mathcal{E}_2 \xrightarrow{\sim} \mathcal{E}_2 + \mathcal{E}_1$

iv) $\epsilon^r_{\mathcal{E}, \mathcal{U}_\mathcal{F}} : \mathcal{E} + \mathcal{O}_{\mathcal{E}, \mathcal{U}_\mathcal{F}} \xrightarrow{\sim} \mathcal{E}$ given by

$$[e, (x, y)] \mapsto e + i_\mathcal{E}(y)$$

v) $\epsilon^l_{\mathcal{E}, \mathcal{U}_\mathcal{F}} : \mathcal{O}_{\mathcal{E}, \mathcal{U}_\mathcal{F}} + \mathcal{E} \xrightarrow{\sim} \mathcal{E}$ given by

$$[(x, y), e] \mapsto e + i_\mathcal{E}(y)$$

vi) $\delta_\mathcal{E} : \mathcal{E} - \mathcal{E} \xrightarrow{\sim} \mathcal{O}_{\mathcal{E}, \mathcal{U}_\mathcal{F}}$ given by

$$[e_1, e_2] \mapsto (k(e_1), y).$$

$k(e_1 - e_2) = 0$

$\Rightarrow i_\mathcal{E}(y) = e_1 - e_2$ ($! \; y \in \mathcal{U}_\mathcal{F}$)

Splittings: Given an extension

$$0 \to \mathcal{G} \xrightarrow{i_\mathcal{E}} \mathcal{E} \xrightarrow{h_\mathcal{E}} \mathcal{F} \to 0 \quad ;$$

$$\mathcal{F} \xleftarrow{s_\mathcal{E}} \mathcal{E}$$

a splitting is $s_\mathcal{E} : \mathcal{F} \to \mathcal{E}$ s.t. $h_\mathcal{E} \circ s_\mathcal{E} = id_\mathcal{F}$.

In this case

$$\phi_s : \mathcal{F} \oplus \mathcal{G} \xrightarrow{\sim} \mathcal{E}$$
$$(x, y) \mapsto (s_\mathcal{E}(x), i_\mathcal{E}(y))$$

Conversely if $\phi : \mathcal{F} \oplus \mathcal{G} \xrightarrow{\sim} \mathcal{E}$ is an isomorphism

then $s_\mathcal{E} : \mathcal{F} \to \mathcal{E}$

$x \mapsto \phi(x, 0)$ is a splitting.

$$\{\text{splittings of } \mathcal{E}\} \xrightarrow{\sim} \{\text{isomorphisms of } \mathcal{F} \oplus \mathcal{G} \xrightarrow{\sim} \mathcal{E}\}$$

Prop 3.6: For two extensions $\mathcal{E}_1$ and $\mathcal{E}_2$:

$\exists$ a bijective correspondence

$$\{\text{splittings of } \mathcal{E}_1 - \mathcal{E}_2\} \longrightarrow \{\text{isomorphisms of } \mathcal{E}_1 \simeq \mathcal{E}_2\}$$

$$\downarrow \wr$$

$$\{\text{isomorphisms } \mathcal{O}_{\mathcal{F},\mathcal{G}} \to \mathcal{E}_1 - \mathcal{E}_2\} \xrightarrow{(i_{\mathcal{E}_1} + s_{\mathcal{E}_2})^*}$$

Part - II : Fix an open cover $\mathcal{U} = \{U_\alpha\}$ of X and extensions of sheaves on each $U_\alpha \cap U_\beta$:

$$0 \to \mathcal{G}|_{X_\alpha \cap X_\beta} \to \mathcal{E}_{\alpha\beta} \to \mathcal{F}|_{U_\alpha \cap U_\beta} \to 0 \quad .$$

s.t. on each $U_\alpha \cap U_\beta \cap U_\gamma$

$$\exists\, F_{\alpha\beta\gamma} : \mathcal{E}_{\alpha\beta} + \mathcal{E}_{\beta\gamma} \xrightarrow{\sim} \mathcal{E}_{\alpha\gamma} \quad \text{satisfying}$$

$$\begin{array}{ccc} \mathcal{E}_{\alpha\beta} + \mathcal{E}_{\beta\gamma} + \mathcal{E}_{\gamma\delta} & \xrightarrow{F_{\alpha\beta\gamma} + id_{\gamma\delta}} & \mathcal{E}_{\alpha\gamma} + \mathcal{E}_{\gamma\delta} \\ \downarrow id_{\mathcal{E}_{\alpha\beta}} + F_{\beta\gamma\delta} & & \downarrow F_{\alpha\gamma\delta} \\ \mathcal{E}_{\alpha\beta} + \mathcal{E}_{\beta\delta} & \xrightarrow[F_{\alpha\beta\delta}]{} & \mathcal{E}_{\alpha\delta} \end{array}$$

in $U_\alpha \cap U_\beta \cap U_\gamma \cap U_\delta$.

The pair $(\{\mathcal{E}_{\alpha\beta}\}, \{F_{\alpha\beta\gamma}\})$ is an extension cocycle.

Given extensions $\mathcal{E}_\alpha$ on each U_α;

$$\partial(\{\mathcal{E}_\alpha\}) = \mathcal{E}_\alpha - \mathcal{E}_\beta \text{ together}$$

$$\begin{array}{ccc} \mathcal{E}_\alpha - \mathcal{E}_\beta + \mathcal{E}_\beta - \mathcal{E}_\gamma & \longrightarrow & \mathcal{E}_\alpha - \mathcal{E}_\gamma \\ \downarrow id_{\mathcal{E}_\alpha} + \delta_{-\mathcal{E}_\beta} + id_{-\mathcal{E}_\gamma} & \nearrow \epsilon_{F,\mathcal{G}} + id_{-\mathcal{E}_\gamma} & \\ \mathcal{E}_\alpha + \mathcal{O}_{F,\mathcal{G}} - \mathcal{E}_\gamma & & \end{array}$$

gives an extension cocycle.

Group of extension classes : {Isomorphism classes of extension cocycles} / {boundary subgroup}

An extension class is said to be a boundary class if it is isomorphic to $d(\{\xi_\alpha\})$.

We denote this group by $\mathbb{E}_G(\mathcal{U}; F, \mathcal{G})$. Given a refinement $\mathcal{U}'$ of $\mathcal{U}$ with a function

$$\rho: \mathcal{U}' \to \mathcal{U} \quad \text{s.t.} \quad U'_\alpha \subseteq U_{\rho(\alpha)} \quad \forall \alpha$$

we get a map $\mathbb{E}_G(\mathcal{U}; F, \mathcal{G}) \to \mathbb{E}_G(\mathcal{U}'; F, \mathcal{G})$ given by

$$(\{\xi_{\alpha\beta}\}, F_{\beta\delta}) \mapsto (\{\xi_{\rho(\alpha)\rho(\beta)}|_{U'_\alpha \cap U'_\beta}\}, F_{\rho(\alpha)\rho(\beta)\rho(\delta)}|_{U'_\alpha \cap U'_\beta \cap U'_\delta})$$

$$\mathbb{E}_G(F, \mathcal{G}) := \varinjlim \mathbb{E}_G(\mathcal{U}; F, \mathcal{G}).$$

Thm 3.13: There is a canonical group isomorphism

$$\mathbb{E}_G(F, \mathcal{G}) \xrightarrow{\sim} \ker\left(\mathrm{Ext}^2_G(F, \mathcal{G}) \to H^0(X, \underline{\mathrm{Ext}}^2_G(F, \mathcal{G}))\right)$$

Here the map in the RHS comes from the Spectral sequence

$$H^i(X, \underline{\mathrm{Ext}}^j_G(F, \mathcal{G})) \Rightarrow \mathrm{Ext}^{i+j}_G(F, \mathcal{G}).$$

$$\| \quad \underline{\mathrm{Ext}}^j_G(F, \mathcal{G})$$

$$R^i\Gamma(X, -) \circ R^j \underline{\mathrm{Hom}}_G(F, -)(\mathcal{G})$$

$$\Rightarrow R^{i+j}\mathrm{Hom}(F, -)(\mathcal{G}).$$

- Choose an inj. object $\mathcal{J}$ s.t. $\mathcal{G} \hookrightarrow \mathcal{J}$ with $\mathcal{J}/\mathcal{G} = \mathcal{Q}$.

$$E^1 = \underline{Ext}^1_{\mathcal{O}}(\mathcal{F}, \mathcal{Q}) \xrightarrow{\sim} \underline{Ext}^2_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) \quad \text{commutes}$$

$$\downarrow \qquad\qquad \downarrow$$

$$E_2^{0,1} = H^0(X, \underline{\mathcal{E}xt}^1_{\mathcal{O}}(\mathcal{F}, \mathcal{Q})) \xrightarrow{\sim} H^0(X, \underline{\mathcal{E}xt}^2_{\mathcal{O}}(\mathcal{F}, \mathcal{G}))$$

- kernel (left side) = kernel (right side).
- we prove, for an open cover $\mathcal{U}$ of X

Lem. 3.14 $\quad \mathbb{E}_{\mathcal{O}}(\mathcal{U}; \mathcal{F}, \mathcal{G}) \xrightarrow{\sim} \check{H}^1(\mathcal{U}, \underline{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{Q}))$.

Taking the direct limit over $\mathcal{U}$ yields

$$\underline{\mathbb{E}}_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) \xrightarrow{\sim} H^1(X, \underline{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{Q}))$$

- Note: Moreover Lem 3.14 will also yield that the group of extension classes does not depend on the refinement function $f: \mathcal{U}' \to \mathcal{U}$.

- Start with an extension cocycle $(\{\mathcal{E}_{\alpha\beta}\}, \mathcal{F}_{\beta\gamma})$. Find $\sigma_{\alpha\beta}$

s.t.
$$\begin{array}{ccc} \mathcal{G} & \hookrightarrow & \mathcal{E}_{\alpha\beta} \quad (\forall\, \alpha, \beta) \\ j \downarrow & \swarrow_{\sigma_{\alpha\beta}} & \\ \mathcal{J} & & \end{array}$$

• Choose $\{\sigma_{\alpha\beta}\}$ in a coherent way i.e. $\forall \alpha, \beta, \gamma$

$$\sigma_{\alpha\beta} + \sigma_{\beta\gamma} = \sigma_{\alpha\gamma} \circ F_{\alpha\beta\gamma} \quad \text{on} \quad U_\alpha \cap U_\beta \cap U_\gamma.$$

$$\zeta_{\alpha\beta\gamma} := (\sigma_{\alpha\beta} + \sigma_{\beta\gamma}) - (\sigma_{\alpha\gamma} \circ F_{\alpha\beta\gamma}) : \mathcal{F} \to \mathcal{J} \quad \text{on} \quad U_\alpha \cap U_\beta \cap U_\gamma$$

$$\begin{cases} (\sigma_{\alpha\beta} + \sigma_{\beta\gamma})\left(i_{\xi_{\alpha\beta} + \xi_{\beta\gamma}}(y)\right) = (\sigma_{\alpha\beta} + \sigma_{\beta\gamma})\left[i_{\xi_{\alpha\beta}}(y), 0\right] \\ \qquad\qquad\qquad\qquad = j \\ (\sigma_{\alpha\gamma} \circ F_{\alpha\beta\gamma})\left(i_{\xi_{\alpha\beta} + \xi_{\beta\gamma}}(y)\right) = j \end{cases}$$

$$\begin{array}{ccc} y & \longmapsto & \xi_{\alpha\beta} + \xi_{\beta\gamma} \\ \| & \ni & \downarrow F_{\alpha\beta\gamma} \\ y & \longmapsto & \xi_{\alpha\gamma} \end{array}$$

$i_{\xi_{\alpha\beta} + \xi_{\beta\gamma}}$, $i_{\xi_{\alpha\gamma}}$

<u>claim</u>: $\zeta_{\alpha\beta\gamma}$ forms a $\check{C}$ech 2 cocycle i.e. on $\forall \alpha, \beta, \gamma, \delta$:

$$\zeta_{\alpha\beta\gamma} - \zeta_{\alpha\beta\delta} + \zeta_{\alpha\gamma\delta} - \zeta_{\beta\gamma\delta} = 0.$$

<u>pf</u>: For x a section of $\mathcal{F}$ pick $e_{\alpha\beta}$, $e_{\beta\gamma}$ and $e_{\gamma\delta}$ from $\xi_{\alpha\beta}$, $\xi_{\beta\gamma}$ and $\xi_{\gamma\delta}$ respectively. Define

$$e_{\alpha\gamma} := F_{\alpha\beta\gamma}([e_{\alpha\beta}, e_{\beta\gamma}]) \; ; \; e_{\beta\delta} := F_{\beta\gamma\delta}([e_{\beta\gamma}, e_{\gamma\delta}])$$

$e_{\alpha\delta} := F_{\alpha\gamma\delta}([e_{\alpha\gamma}, e_{\gamma\delta}])$.

Note $e_{\alpha\delta} = F_{\alpha\beta\delta}([e_{\alpha\beta}, e_{\beta\delta}]) = F_{\alpha\gamma\delta}([e_{\alpha\gamma}, e_{\gamma\delta}])$.

$\therefore (\tau_{\alpha\beta\gamma} + \tau_{\alpha\gamma\delta} - \tau_{\alpha\beta\delta} - \tau_{\beta\gamma\delta})(x)$ $\quad e_{\alpha\gamma}$

$= \sigma_{\alpha\beta}(e_{\alpha\beta}) + \sigma_{\beta\gamma}(e_{\beta\gamma}) - \sigma_{\alpha\gamma}(F_{\alpha\beta\gamma}[e_{\alpha\beta}, e_{\beta\gamma}])$

$+ \sigma_{\alpha\gamma}(e_{\alpha\gamma}) + \sigma_{\gamma\delta}(e_{\gamma\delta}) - \sigma_{\alpha\delta}(F_{\alpha\gamma\delta}[e_{\alpha\gamma}, e_{\gamma\delta}])$ $\quad = e_{\alpha\delta}$

$- \sigma_{\alpha\beta}(e_{\alpha\beta}) - \sigma_{\beta\delta}(e_{\beta\delta}) + \sigma_{\alpha\delta}(F_{\alpha\beta\delta}[e_{\alpha\beta}, e_{\beta\delta}])$ $\quad = e_{\alpha\gamma}$

$- \sigma_{\beta\gamma}(e_{\beta\gamma}) - \sigma_{\gamma\delta}(e_{\gamma\delta}) + \sigma_{\beta\gamma}(F_{\beta\gamma\delta}[e_{\beta\gamma}, e_{\gamma\delta}]) = 0.$

$\quad = e_{\beta\gamma}$

So $[\{\tau_{\alpha\beta\gamma}\}] \in \check{H}^2(\mathcal{U}, \underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{I}))$

$\because \mathcal{I}$ is injective $\Rightarrow \check{H}^2(\mathcal{U}, \underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{I})) = 0$

$\Rightarrow \exists\ \tau_{\alpha\beta}$ of $\underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{I})$ 1 cochain s.t.

$$\tau_{\alpha\beta} + \tau_{\beta\gamma} - \tau_{\alpha\gamma} = \tau_{\alpha\beta\gamma}.$$

Modify $\widetilde{\sigma}_{\alpha\beta} := \sigma_{\alpha\beta} - \tau_{\alpha\beta} \circ k_{\mathcal{E}_{\alpha\beta}}$

Then $(\widetilde{\sigma}_{\alpha\beta} + \widetilde{\sigma}_{\beta\gamma})([e_{\alpha\beta}, e_{\beta\gamma}])$

$= \widetilde{\sigma}_{\alpha\beta}(e_{\alpha\beta}) + \widetilde{\sigma}_{\beta\gamma}(e_{\beta\gamma})$

$$= \sigma_{\alpha\beta}(e_{\alpha\beta}) - \tau_{\alpha\beta} \circ k_{\varepsilon_{\alpha\beta}}(e_{\alpha\beta}) + \sigma_{\beta\gamma}(e_{\beta\gamma}) - \tau_{\beta\gamma} \circ k_{\varepsilon_{\beta\gamma}}(e_{\beta\gamma})$$

$$= \sigma_{\alpha\gamma} \circ F_{\alpha\beta\gamma}([e_{\alpha\beta}, e_{\beta\gamma}]) + \tau_{\alpha\beta\gamma}([e_{\alpha\beta}, e_{\beta\gamma}])$$

$$- \tau_{\alpha\beta} \circ k_{\varepsilon_{\alpha\beta}}(e_{\alpha\beta}) - \tau_{\beta\gamma} k_{\varepsilon_{\beta\gamma}}(e_{\beta\gamma})$$

$$= \sigma_{\alpha\gamma} \circ F_{\alpha\beta\gamma}([e_{\alpha\beta}, e_{\beta\gamma}]) - \tau_{\alpha\gamma} \circ F_{\alpha\beta\gamma}([e_{\alpha\beta}, e_{\beta\gamma}])$$

$$= \tilde{\sigma}_{\alpha\gamma} \circ F_{\alpha\beta\gamma} .$$

Given an extension cocycle $(\{\varepsilon_{\alpha\beta}\}, \{F_{\alpha\beta\gamma}\})$ and functions $\{\sigma_{\alpha\beta}\}$; we get $\eta_{\alpha\beta} : \mathcal{F} \to \mathcal{Q}$ satisfying

$$\eta_{\alpha\beta} + \eta_{\beta\gamma} = \eta_{\alpha\gamma}$$

i.e. an element $\hat{w}$ of $\check{H}^1(\mathcal{U}, \underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{Q}))$.

- if $\{\sigma'_{\alpha\beta}\}$ is another function that corresponds to $w' \in \check{H}^1(\mathcal{U}, \underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{Q}))$ then $\{\sigma_{\alpha\beta} - \sigma'_{\alpha\beta}\}$ is a cocycle with values in $\underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{A})$ which corresponds to $w - w' \in \check{H}^1(\mathcal{U}, \underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{A})) = 0$

$$\Rightarrow \boxed{w = w'}.$$

injectivity :

Let $(\{\mathcal{E}_{\alpha\beta}\}, F_{\alpha\beta\gamma})$ be a cocycle which corresponds to $0 \in \check{H}^1(\mathcal{U}, \underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{Q}))$. $\exists$ a 0-cochain $\{\eta_\alpha\}$ s.t.

$$\eta_{\alpha\beta} = \eta_\beta - \eta_\alpha \quad \text{for all } \alpha, \beta. \qquad \eta_\alpha : \mathcal{F} \to \mathcal{Q}$$

$$\begin{array}{ccccccc} 0 & \to & \mathcal{K} & \to & \mathcal{E}_{\alpha\beta} & \to & \mathcal{F} \to 0 \\ & & \| & & \downarrow \sigma_{\alpha\beta} & & \downarrow \eta_{\alpha\beta} \\ 0 & \to & \mathcal{K} & \to & \mathcal{J} & \to & \mathcal{Q} \to 0 \end{array}$$

Let $\mathcal{E}_\alpha := \eta_\alpha^* \mathcal{J}$. Then $\eta_{\alpha\beta}^* \mathcal{J} = (\eta_\beta - \eta_\alpha)^* \mathcal{J}$

$$\simeq \eta_\beta^* \mathcal{J} - \eta_\alpha^* \mathcal{J}$$

$$= \mathcal{E}_\beta - \mathcal{E}_\alpha = \partial(\{\mathcal{E}_\alpha\}).$$

by universal prop of the pullback ;

$$\eta_{\alpha\beta}^* \mathcal{J} \simeq \mathcal{E}_{\alpha\beta}.$$

Hence; $(\{\mathcal{E}_{\alpha\beta}\}, F_{\alpha\beta\gamma})$ is isomorphic to a boundary class.

Surjectivity: Let $w \in \check{H}^1(\mathcal{U}; \underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{Q}))$

$$[\eta_{\alpha\beta}].$$

Let $\xi_{\alpha\beta} := \eta^*_{\alpha\beta} 1$ with

$$\tau_{\alpha\beta\gamma} : \xi_{\alpha\beta} + \xi_{\beta\gamma} = (\eta^*_{\alpha\beta} + \eta^*_{\beta\gamma}) 1 \simeq (\eta_{\alpha\beta} + \eta_{\beta\gamma})^* 1$$

$$= \eta^*_{\alpha\gamma} 1 = \xi_{\alpha\gamma}.$$

Thus we have

$$\underline{\mathbb{E}}_{\mathcal{O}}(\mathcal{U}; \mathcal{F}, \mathcal{G}) \xrightarrow{\sim} \check{H}^1(\mathcal{U}, \underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{Q})).$$

Moreover it is compatible with refinements. Taking the direct limit gives

$$\underline{\mathbb{E}}_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) \xrightarrow{\sim} H^1(X, \underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}, \mathcal{Q})).$$

Part - III : Generalization :

$\mathcal{F} \rightsquigarrow \mathcal{F}^\bullet$ (complex of sheaves of $\mathcal{O}_X$-modules)

$\mathcal{G} \rightsquigarrow \mathcal{G}$ (an one object-complex at deg. 0)

Thm :

- $\mathbb{E}_{\mathcal{O}}(\mathcal{F}^\bullet, \mathcal{G}) \simeq \ker\left(\mathrm{Ext}^2(\mathcal{F}^\bullet, \mathcal{G}) \to H^0(X, \underline{\mathcal{E}\mathrm{xt}}^2_{\mathcal{O}}(\mathcal{F}^\bullet, \mathcal{G}))\right)$

Def : Fix an injective sheaf $\mathcal{I}$ s.t $\mathcal{G} \hookrightarrow \mathcal{I}$.

$$\mathrm{Ext}^n_{\mathcal{O}}(\mathcal{F}^\bullet, \mathcal{G}) := \mathrm{Hom}_{\mathcal{O}}(\mathcal{F}^\bullet, \mathcal{I}^\bullet[n]) / \{\text{chain homotopy}\}$$

Def : An extension $(\mathcal{E}^\bullet, i, k)$ of $\mathcal{F}^\bullet$ by $\mathcal{G}^\bullet$ is a complex $\mathcal{E}^\bullet$ together with homomorphism of complexes

$i : \mathcal{G}^\bullet \to \mathcal{E}^\bullet$ and $k : \mathcal{E}^\bullet \to \mathcal{F}^\bullet$ that fits exact seq.

$$0 \to \mathcal{G}^n \to \mathcal{E}^n \to \mathcal{F}^n \to 0 \quad (\text{for all } n \in \mathbb{Z}).$$

→ Take an extension $\mathcal{E}^\bullet$ of $\mathcal{F}^\bullet$ by $\mathcal{G}$.

The embedding $\mathcal{G} \hookrightarrow \mathcal{I}^\bullet$ extends to a homomorphism of graded sheaves $\Phi^\bullet : \mathcal{E}^\bullet \to \mathcal{I}^\bullet$. The homomorphism of complexes $\partial_{\mathcal{I}^\bullet}\Phi^\bullet - \Phi^\bullet\partial_{\mathcal{E}^\bullet} : \mathcal{E}^\bullet \to \mathcal{I}^\bullet[1]$ sends $\mathcal{G}$ to 0, and hence induces $\psi : \mathcal{F}^\bullet \to \mathcal{I}^\bullet[1]$. We associate the

class $[\Psi] \in \mathrm{Ext}^1_{\mathcal{O}}(F^\bullet, \mathcal{G})$.

Remark: This map is neither surjective nor injective. However if $F^i = 0 \; ; \forall i > 1$ then this is a bijection.

$$\left(\partial_{I^0} \Phi^0 \circ \Phi^1 \partial_{\mathcal{E}^0}\right)\left(i_{\mathcal{E}^0}(u)\right)$$

$$= \partial_{I^0} \Phi^0 \left(i_{\mathcal{E}^0}(y)\right)$$

$$= \partial_{I^0}(j(y)) = 0.$$

$$\begin{array}{ccccc}
0 & & & & \\
\downarrow & & & & \\
\mathcal{G} & \xrightarrow{j} & I^0 & & \\
{\scriptstyle i_{\mathcal{E}^0}}\downarrow & \nearrow & & & \\
\mathcal{E}^0 & & \xrightarrow{\Phi^0} & & \\
0 & & & & \\
\downarrow & & & & \\
\mathcal{G} & \xrightarrow{j} & I^0 & & \\
\downarrow & & \downarrow{\scriptstyle \partial_{I^0}} & & \\
0 & \to & I^1 & & \\
\downarrow & & & & \\
0 & & & &
\end{array}$$

Lem 3.18: $\exists$ a canonical isomorphism

$$H^1(X, \underline{\mathrm{Hom}}_{\mathcal{O}}(F^\bullet, \mathcal{G}^\bullet[n])) \xrightarrow{\sim} \ker\left(\mathrm{Ext}^{n+1}_{\mathcal{O}}(F^\bullet, \mathcal{G}) \to H^0(X, \underline{\mathrm{Ext}}^{n+1}_{\mathcal{O}}(F^\bullet, \mathcal{G}))\right)$$

Lem 3.19: $\mathbb{E}_{\mathcal{O}}(\mathcal{U}; F, \mathcal{G}) \xrightarrow{\sim} \check{H}^1(\mathcal{U}, \underline{\mathrm{Hom}}_{\mathcal{O}}(F^\bullet, \mathcal{G}^\bullet[1]))$.

{ Together these two lemmas will prove

$$\mathbb{E}_{\mathcal{O}}(F, \mathcal{G}) \xrightarrow{\sim} \ker\left(\mathrm{Ext}^2_{\mathcal{O}}(F^\bullet, \mathcal{G}) \to H^0(X, \underline{\mathrm{Ext}}^2_{\mathcal{O}}(F^\bullet, \mathcal{G}))\right)$$

pf (Lem 3.11) : Pick $(\{\mathcal{E}^{\bullet}_{\alpha\beta}\}, F^{\bullet}_{\alpha\beta\gamma})$ an extension cocycle and choose $\sigma^{\bullet}_{\alpha\beta} : \mathcal{E}^{\bullet}_{\alpha\beta} \to \mathcal{J}^{\bullet}$ satisfying the cocycle condition. This is a map of graded sheaves. However

$$\sigma^{\bullet}_{\alpha\beta} \circ \partial_{\mathcal{E}^{\bullet}_{\alpha\beta}} - \partial_{\mathcal{I}^{\bullet}} \circ \sigma^{\bullet}_{\alpha\beta} : \mathcal{E}^{\bullet}_{\alpha\beta} \to \mathcal{J}^{\bullet}[1]$$ sends $\mathcal{G}$ to 0

and induces a homomorphism of chain complexes

$$\psi : \mathcal{F}^{\bullet} \to \mathcal{J}^{\bullet}[1].$$

Rest of the proof goes with the obvious modifications in the Thm 3.13.

pf (Lem 3.12) : $A := \ker\left(\underline{\mathrm{Ext}}^{n+1}_{\mathcal{O}}(\mathcal{F}^{\bullet}, \mathcal{G}) \to H^0(X, \underline{\mathcal{E}xt}^{n}_{\mathcal{O}}(\mathcal{F}^{\bullet}, \mathcal{G}))\right)$.

Define $A \to H^1(X, \underline{\mathrm{Hom}}(\mathcal{F}^{\bullet}, \mathcal{J}^{\bullet}[n]))$ as follows

Take a class from A rep. by

$\phi^{\bullet} : \mathcal{F}^{\bullet} \to \mathcal{J}^{\bullet}[n+1]$ which vanishes in

$$\underset{\text{in}}{\mathcal{Q}} \qquad\qquad H^0(X, \underline{\mathcal{E}xt}^{n}_{\mathcal{O}}(\mathcal{F}^{\bullet}, \mathcal{G}))$$

so $\exists$ an open cover $\{U_\alpha\}$ of X and $\phi_\alpha : \mathcal{F}^{\bullet}|_{U_\alpha} \to \mathcal{J}^{\bullet}[n]|_{U_\alpha}$

s.t. $\phi|_{U_\alpha} = \partial_{\mathcal{I}^{\bullet}} \circ \phi_\alpha - \phi_\alpha \circ \partial_{\mathcal{F}^{\bullet}}$.

Define $\phi_{\alpha\beta} := \phi_\alpha - \phi_\beta$. This defines 1 Čech cocycle with values in $\underline{\mathrm{Hom}}_{\mathcal{O}}(\mathcal{F}^{\bullet}, \mathcal{J}^{\bullet}[n])$. Thus we get an injective

map $A \to \check{H}^1(\mathcal{U}, \underline{\mathrm{Hom}}_G(F^\bullet, J^\bullet[n]))$.

Conversely: if $[\{\phi_{\alpha\beta}\}] \in \check{H}^1(X, \underline{\mathrm{Hom}}_G(F^\bullet, J^\bullet[n]))$ be an element wrt open cover $\mathcal{U} = \{U_\alpha\}_\alpha$

$\underline{\mathrm{Hom}}^0_G(F^\bullet, J^\bullet[n]) := \bigoplus_{k\in\mathbb{Z}} \mathrm{Hom}(F^k, J^{n+k})$ · is a finite sum of flabby sheaves

and thus $\exists \{\phi_\alpha\}$ s.t. $\phi_{\alpha\beta} = \phi_\beta - \phi_\alpha$.

(map of graded sheaves)

$\hookrightarrow \phi_\alpha : F^\bullet|_{U_\alpha} \to J^\bullet[n]|_{U_\alpha}$

$\psi_\alpha := \phi_\alpha \circ \partial_F - \partial_I \circ \phi_\alpha$ agrees on $U_\alpha \cap U_\beta$

$\psi_\alpha - \psi_\beta = (\phi_\alpha - \phi_\beta)\circ\partial_F - \partial_I \circ (\phi_\alpha - \phi_\beta)$

$= \phi_{\alpha\beta}\partial_F - \partial_I \phi_{\alpha\beta} = 0$. and glues to get

$\psi : F^\bullet \to J^\bullet[n+1]$ and hence an element

$[\psi] \in \mathrm{Ext}^{n+1}_G(F^\bullet, G)$ which restricts to

$\psi|_{U_\alpha} = \psi_\alpha = \phi_\alpha \circ \partial_F - \partial_I \circ \phi_\alpha$

$\in H^0(X, \underline{\mathrm{End}}^{n+1}_G(F^\bullet, G))$. ∎

• Spectral sequence and localization map :

$$E_2^{p,q} = H^p(X, \underline{Ext}^q_{\mathcal{O}}(\mathcal{F}, \mathcal{G})) \Rightarrow Ext^{p+q}_{\mathcal{O}}(\mathcal{F}, \mathcal{G}).$$

→ Five term exact sequence :

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d} E_2^{2,0} \to H^2$$

on page 2 of spectral sequence

$(p,q) \mapsto (p+2, q-1)$

on page 2

$$E_2^{p,q} = \frac{\ker(E_1^{p,q} \to E_1^{p+2,q-1})}{\mathrm{Im}(E_1^{p-2,q+1} \to E_1^{p,q})}$$

→ $\exists$ a filtration

$$F^i H^n \supseteq F^{i-1} H^n \supseteq \dots \supseteq F^1 H^n \supseteq F^0 H^n = H^n$$

s.t. $\dfrac{F^i H^n}{F^{i+1} H^n} \simeq E_\infty^{i,n-i}$

$$Z_{r+1}^{p,q} = \ker(E_r^{p,q} \to E_r^{p+r, q-r+1})$$

$$B_{r+1} = \mathrm{Im}(E_r^{p-r, q+r-1} \to E_r^{p,q})$$

$$B_\infty^{p,q} = \bigcup B_r^{p,q}$$

$$Z_\infty^{p,q} = \bigcap Z_r^{p,q}$$

$$E_{r+1}^{p,q} := \frac{Z_{r+1}^{p,q}}{B_{r+1}}$$

$$E_\infty^{p,q} := \frac{Z_\infty^{p,q}}{B_\infty^{p,q}}$$

$$F^2H^2 \supseteq F^1H^2 \supseteq F^0H^2 = H^2$$

$$E_{r+1}^{0,2} = \ker\left(E_r^{0,2} \to E_r^{r,3-r}\right)$$

- $H^2 \twoheadrightarrow \dfrac{H^2}{F^1H^2} \simeq E_\infty^{0,2} \hookrightarrow E_2^{0,2}$ so $E_{r+1}^{0,2} \subseteq E_r^{0,2}$ $(\forall r)$

- In general:

$$H^{n+1} \twoheadrightarrow \frac{H^{n+1}}{F^1H^{n+1}} \simeq E_\infty^{0,n+1} \hookrightarrow E_2^{0,n+1}$$

$\underline{\mathcal{E}xt}_{\mathcal{O}}^n(\mathcal{F}^\bullet, \mathcal{G})$ sheaf:

Sheafification of

$$U \mapsto \mathcal{E}xt_{\mathcal{O}_U}^n(\mathcal{F}^\bullet|_U, \mathcal{G}|_U)$$

Fix an injective resolution $\mathcal{G} \hookrightarrow \mathcal{J}^\bullet$

$$\underline{\mathcal{E}xt}_{\mathcal{O}}^n(\mathcal{F}^\bullet, \mathcal{G}) = \mathcal{H}^n\left(\underline{\mathcal{H}om}_{\mathcal{O}}^\bullet(\mathcal{F}^\bullet, \mathcal{J}^\bullet[n])\right)$$

where $\underline{\mathcal{H}om}_{\mathcal{O}}^k(\mathcal{F}^\bullet, \mathcal{J}^\bullet[n]) = \bigoplus_i \mathcal{H}om\left(\mathcal{F}^i, \mathcal{J}^{i+k+n}\right)$

Čech Cohomology:

$\mathcal{U}, \mathcal{V}: \mathcal{U}' \to \mathcal{U}$

$\rho_1^*: \check{H}^1(\mathcal{U}, \mathcal{F}) \to \check{H}^1(\mathcal{U}', \mathcal{F})$ $\quad U'_\alpha \subseteq U_{\rho'(\alpha)}$, $U'_\alpha \subseteq U_{\rho^2(\alpha)}$

$\phi = [\{\phi_{\alpha\beta}\}] \mapsto (\rho_1^* \phi)_{\alpha\beta} : U'_\alpha \cap U'_\beta \to$

$\| \quad \phi_{\rho'(\alpha)\rho'(\beta)}|_{U'_\alpha \cap U'_\beta}$

$(\rho_2^* \phi)_{\alpha\beta} = \phi_{\rho^2(\alpha)\rho^2(\beta)}|_{U'_\alpha \cap U'_\beta}$

$\phi_{\rho'(\alpha)\rho^2(\alpha)} + \phi_{\rho^2(\alpha)\rho^2(\beta)} = \phi_{\rho'(\alpha)\rho^2(\beta)}$

$|_{U'_\alpha \cap U'_\beta}$ $\quad \phi_{\rho'(\alpha)\rho'(\beta)} + \phi_{\rho'(\beta)\rho^2(\beta)} = \phi_{\rho'(\alpha)\rho^2(\beta)}$

$\therefore \psi_\alpha + (\rho_2^* \psi)_{\alpha\beta} = \phi_{\rho'(\alpha)\rho^2(\beta)}|_{U'_\alpha \cap U'_\beta}$

$(\rho_1^* \phi)_{\alpha\beta} + \psi_\beta = \phi_{\rho'(\alpha)\rho^2(\beta)}|_{U'_\alpha \cap U'_\beta}$

$\Rightarrow (\rho_2^* \phi)_{\alpha\beta} - (\rho_1^* \phi)_{\alpha\beta} = \psi_\beta - \psi_\alpha$

$\Rightarrow$ independent of the embedding ρ', ρ^2.

$$\begin{array}{ccc} \mathbb{E}(\mathcal{U}; F, G) & \xrightarrow{\sim} & \check{H}^1(\mathcal{U}, \mathcal{H}om_{\mathcal{O}}(F, G)) \\ \downarrow & \supseteq & \downarrow \\ \mathbb{E}(\mathcal{U}', F, G) & \xrightarrow{\sim} & \check{H}^1(\mathcal{U}', \mathcal{H}om_{\mathcal{O}}(F, G)) \end{array}$$

Rmk Sum of the extensions is indeed the sum in $\overline{\mathrm{Ext}^1_{\mathcal{O}}(F, G)}$:

- $a, b \in Ext^1_G(F, \mathcal{O}_Y)$ $\leftrightarrow$ $0 \to \mathcal{O}_Y \to E_a \to F \to 0$

$a : F \to I/\mathcal{O}_Y$ $\qquad 0 \to \mathcal{O}_Y \to E_b \to F \to 0$

where $\quad 0 \to \mathcal{O}_Y \to E_a \to F \to 0$

$\qquad\qquad \| \quad \downarrow \quad \downarrow$

$0 \to \mathcal{O}_Y \to \mathcal{I} \to \mathcal{I}/\mathcal{O}_Y \to 0$

$$E_a = \{(x, i) \in F \oplus \mathcal{I} \mid i \pmod{\mathcal{O}_Y} = a(x)\}$$

$$E_b = \{(x, i) \in F \oplus \mathcal{I} \mid i \pmod{\mathcal{O}_Y} = b(x)\}$$

$$\therefore E_{a+b} = \{(x, i) \in F \oplus \mathcal{I} \mid i \pmod{\mathcal{O}_Y} = (a+b)(x)\}$$

$$E_a \boxplus E_b := (+)_* \Delta^* (E_a \oplus E_b)$$

$$\| \quad \{((x,i),(y,j)) \mid x = y\}$$

$$= \{(x, i, j) \mid i \pmod{\mathcal{O}_Y} = a(x), \; j \pmod{\mathcal{O}_Y} = b(x)\}$$

$(g_1, g_2) \mapsto ((0, g_1), (0, g_2))$

$\mathcal{O}_Y \oplus \mathcal{O}_Y \to \Delta^*(E_a \oplus E_b)$

$(+) \downarrow \qquad \downarrow$

$\mathcal{O}_Y \to E_a \boxplus E_b$

$g_1 + g_2$

$\alpha(x, i, j) := \{(x, i+j)\}$

$\beta(g) := \{(0, g)\}$

$\exists! \; \vdots \; E_{a+b}$

$$d\,(0, g_1, g_2) = (0,\ g_1 + g_2)$$

Example: $\mathcal{F}^\bullet$: complex $\qquad R = \dfrac{k[\varepsilon]}{(\varepsilon^2)}$

$$\cdots \to R \xrightarrow{\varepsilon} R \xrightarrow{\varepsilon} R \to \cdots$$

$$d_{\mathcal{F}} = k[0]$$

$$\mathcal{E}^\bullet = \begin{cases} R & n \neq 0 \\ k \oplus R & n = 0 \end{cases}$$

with $d_{\mathcal{E}}^n = id \quad (n \neq -1, 0)$

$$d_{\mathcal{E}}^{-1}(r) = (a, a\varepsilon) \qquad d_{\mathcal{E}}^0(x, a + b\varepsilon) = \varepsilon a$$

$a + b\varepsilon$

$$\therefore d_{\mathcal{E}}^0 d_{\mathcal{E}}^{-1}(r) = d_{\mathcal{E}}^0(a, a\varepsilon) = 0.$$

if $s^\bullet : \mathcal{F}^\bullet \to \mathcal{E}^\bullet$ a section then

$$s^{-1} = id, \quad s^1 = id$$

$$\begin{array}{ccc} R & \xrightarrow{s^{-1}} & R \\ d_{\mathcal{E}}^{-1} \downarrow & & \downarrow s^0 \\ k \oplus R & \xrightarrow{\varepsilon} & R \end{array}$$

$$s^0(f) = (h(f), f).$$

$h : R \to k$ (R-linear)

$$s^0(\varepsilon) = (h(\varepsilon), \varepsilon)$$

$$\| \qquad\qquad \| \qquad\qquad \|$$

$$d_{\mathcal{E}}^{-1}(1) = (1, \varepsilon)$$

$\Rightarrow h(\xi) = 1$

$\Rightarrow \xi h(1) = 1$

$\xi \neq 0$

$h = R/(\xi)$.